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# MATHEMATICS

## EXTENSION 2

ASSESSMENT TASK NO: 3

Date : June 12, 2014

**TIME ALLOWED:** *Reading time: 5 minutes*  
*Working time: 1 hour*

**INSTRUCTIONS:** This paper consists of 3 questions  
ANSWER all 3 questions and Multiple Choice questions  
Start each question in a new booklet  
Marks may not be given for untidy and poorly arranged work.  
All relevant working should be shown.

***THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM***

**OUTCOMES ASSESSED:**

|    |                                                                                                                                    |
|----|------------------------------------------------------------------------------------------------------------------------------------|
| E7 | Uses the techniques of slicing and cylindrical shells to determine volumes                                                         |
| E8 | Applies further techniques of integration, including partial fractions, integration by parts and recurrence formulae, to problems. |
| E9 | Communicates abstract ideas and relationships using appropriate notation and logical argument.                                     |

|                  | MARKS |
|------------------|-------|
| QUESTION 1 Topic | /13   |
| QUESTION 2 Topic | /11   |
| QUESTION 3 Topic | /14   |
| Multiple Choice: | /4    |
|                  | /42   |

*This assessment task constitutes 20% of the Higher School Certificate Course Assessment.*

## Section I

4 marks

### Attempt Questions 1- 4

Use the multiple-choice answer sheet for Questions 1 – 4.

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1. The antiderivative of  $\frac{e^{2x}}{2e^{2x}-1}$  ( for  $e^{2x} > \frac{1}{2}$ ) is

(A)  $4\log_e(2e^{2x}-1)$

(B)  $2\log_e(2e^{2x}-1)$

(C)  $\frac{1}{2}\log_e(2e^{2x}-1)$

(D)  $\frac{1}{4}\log_e(2e^{2x}-1)$

2. Using a suitable substitution,  $\int_0^{\frac{\pi}{3}} \sin^3(x)\cos^4(x)dx$  can be expressed in terms of  $u$  as

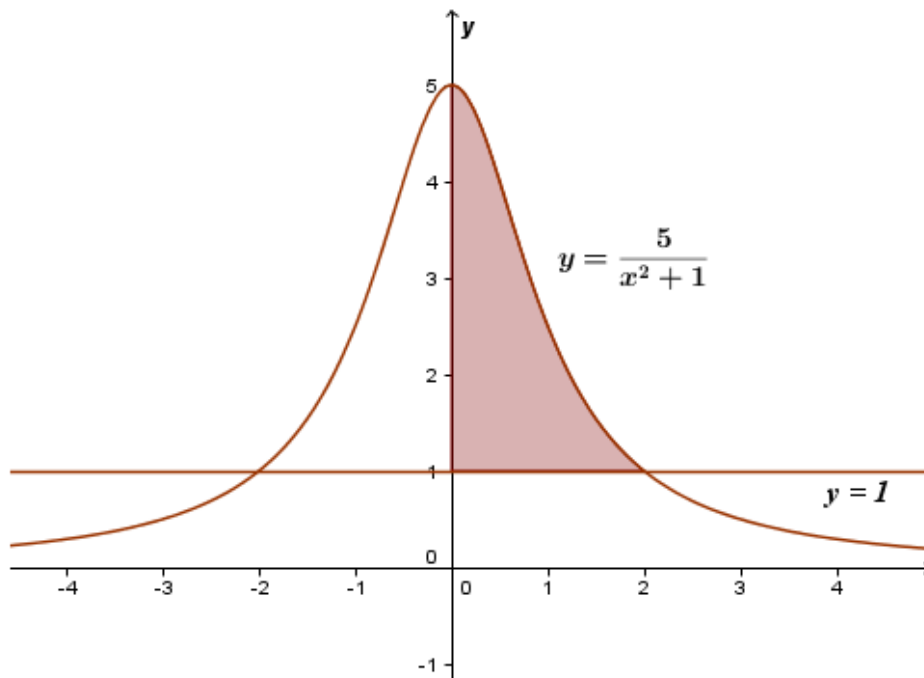
(A)  $\int_{\frac{1}{2}}^1 (u^6 - u^4) du$

(B)  $\int_1^{\frac{1}{2}} (u^6 - u^4) du$

(C)  $\int_0^{\frac{\sqrt{3}}{2}} (u^6 - u^4) du$

(D)  $\int_0^{\frac{\sqrt{3}}{2}} (u^4 - u^6) du$

3. The graph of  $f(x) = \frac{5}{x^2 + 1}$  for  $x \geq 0$ , is shown below. The shaded region is bounded by the graph of  $f(x)$ , the  $y$ -axis and the line with equation  $y = 1$ .



The shaded region is rotated about the  $x$ -axis to form a solid of revolution.

The volume of the solid, in cubic units, is given by

(A)  $\pi \int_0^2 \left( \left( \frac{5}{x^2 + 1} \right)^2 - 1 \right) dx$

(B)  $\pi \int_0^2 \left( \frac{5}{x^2 + 1} - 1 \right)^2 dx$

(C)  $\pi \int_0^2 \left( \frac{5}{x^2 + 1} \right)^2 dx$

(D)  $\pi \int_0^2 \left( \frac{5}{x^2 + 1} - 1 \right) dx$

4. The area bounded by the function  $f(x) = \sqrt{x^2 - 9}$ , the  $x$ -axis and the line with equation  $x = 5$  is rotated about the  $y$ -axis to form a solid of revolution.

The volume of this solid in cubic units is given by:

(A)  $\pi \int_0^4 (y^2 - 9) dy$

(B)  $\pi \int_0^4 (y^2 + 9) dy$

(C)  $\pi \int_0^4 (16 - y^2) dy$

(D)  $\pi \int_0^4 \left(5 - \sqrt{y^2 + 9}\right)^2 dy$

## Section II

### Attempt Question 1- 3

Answer each question on a SEPARATE writing booklet.

Your response should include relevant mathematical reasoning and/or calculations.

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**Question 1** ( 13 marks ) Use a SEPARATE writing booklet.

a) (i) Differentiate  $\frac{\log_e x}{x}$ . 1

(ii) Hence, evaluate  $\int_e^{e^2} \frac{1 - \log_e x}{x \log_e x} dx$  in simplest form. 3

b) Use a suitable substitution to evaluate  $\int_3^8 \frac{2x}{(x-1) + \sqrt{x+1}} dx$  4

c) (i) Find the real numbers  $A$ ,  $B$  and  $C$  such that

$$\frac{2x^2 + 7x - 1}{(x-2)(x^2 + x + 1)} = \frac{A}{x-2} + \frac{Bx+C}{x^2 + x + 1} \quad \text{2}$$

(ii) Hence, find:

$$\int \frac{2x^2 + 7x - 1}{(x-2)(x^2 + x + 1)} dx \quad \text{3}$$

**End of Question 1**

**Question 2** ( 11 marks ) Use a SEPARATE writing booklet.

- a) Using the substitution  $t = \tan \frac{x}{2}$ , or otherwise , **3**

evaluate  $\int_0^{\frac{\pi}{2}} \frac{1}{2\sin x + \cos x + 3} dx$ .

b) Let  $I_n = \int_0^1 (1-x^2)^n dx$  and  $J_n = \int_0^1 x^2 (1-x^2)^n dx$

- (i) Applying integration by parts to  $I_n$ , show that  $I_n = 2n J_{n-1}$ . **3**

- (ii) Hence, show that  $I_n = \frac{2n}{2n+1} I_{n-1}$ . **2**

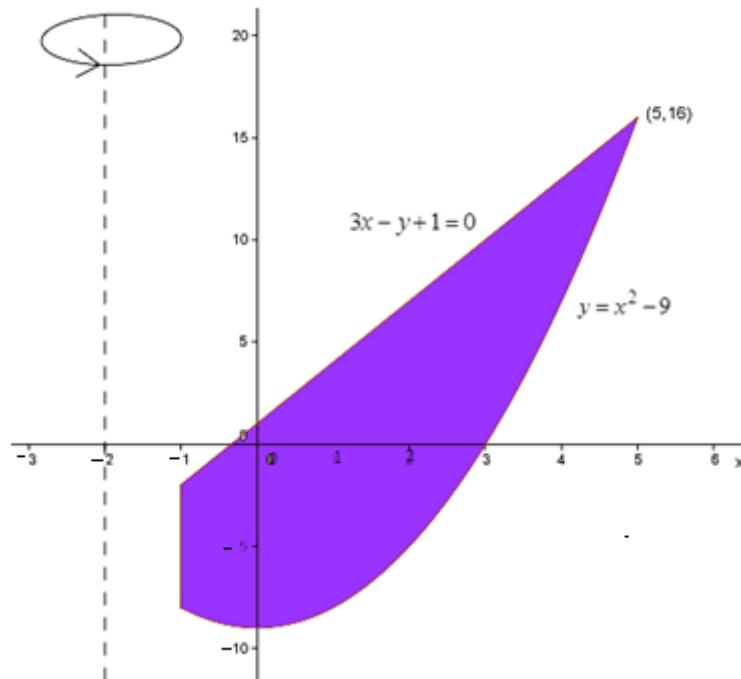
- (iii) Show that  $J_n = I_n - I_{n+1}$ , and hence deduce that  $J_n = \frac{1}{2n+3} I_n$ . **2**

- (iv) Hence, write down a reduction formula for  $J_n$  in terms of  $J_{n-1}$ . **1**

**End of Question 2**

**Question 3** ( 14 marks ) Use a SEPARATE writing booklet.

- a) The region bounded by the curve  $y = x^2 - 9$ , and the line  $3x - y + 1 = 0$  and the line  $x = -1$  is rotated about the line  $x = -2$  to form a solid.



- (i) Using method of cylindrical shells, show that the volume of an elemental shell is given by

**3**

$$\delta V = 2\pi(x+2)(10+3x-x^2)\delta x$$

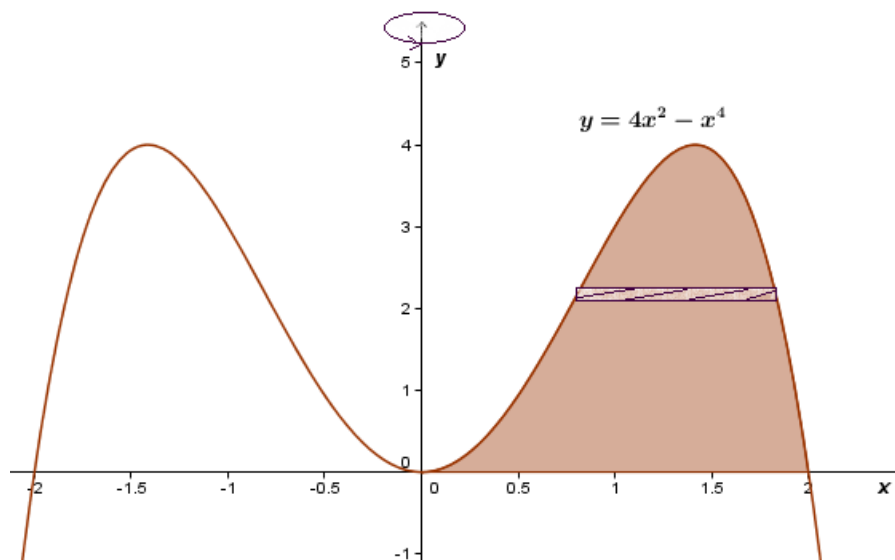
- (ii) Find the volume of the solid formed.

**2**

**Question 3 continues on next page**

- b) The shaded region shown below represents the area bounded by the curve

$y = 4x^2 - x^4$ , the  $x$ -axis for  $0 \leq x \leq 2$ .

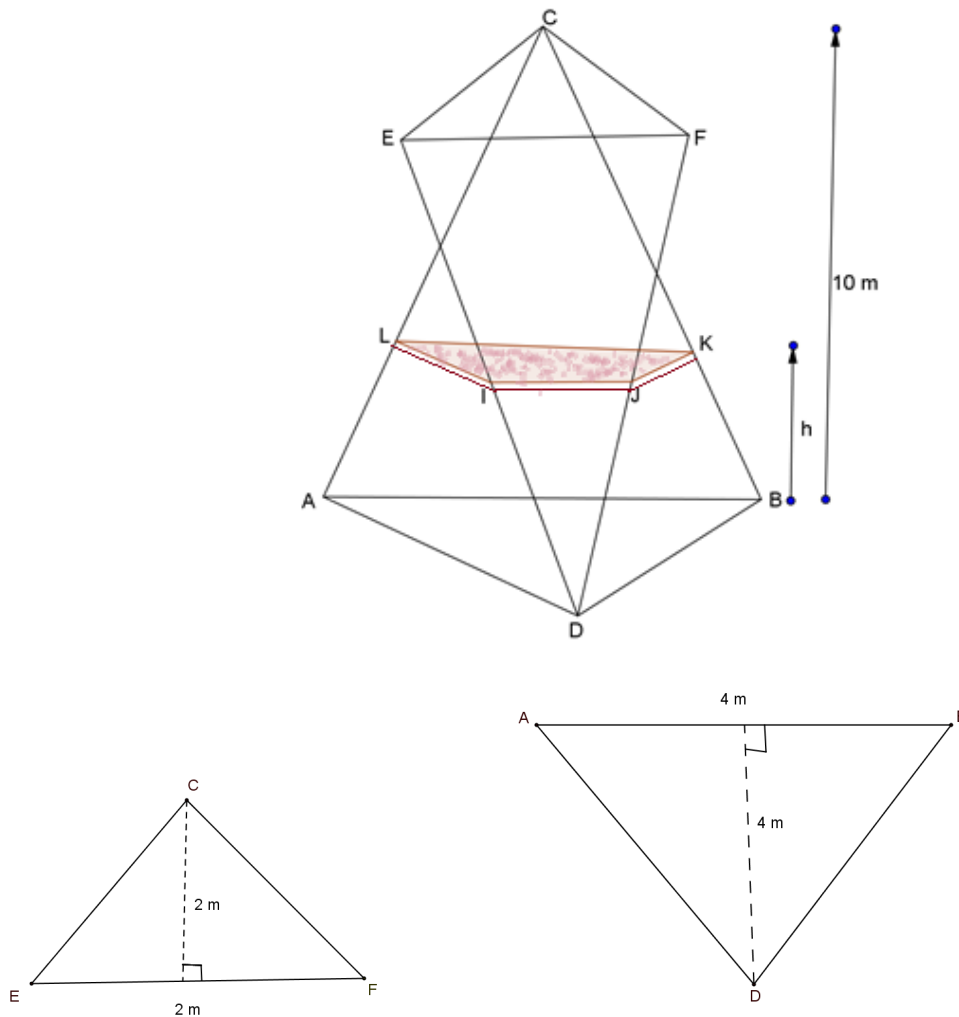


By taking slices perpendicular to the axis of rotation, use the method of slicing to find the volume of the solid obtained by rotating the shaded region about the  $y$ -axis?

**3**

**Question 3 continues on next page**

- c) A large sculpture has a triangular base ( $ABD$ ) and top ( $EFC$ ) as shown. The height of the sculpture is 10 metres. Each cross-section parallel to the base is an isosceles trapezium, and all other dimensions are shown in the diagram below. A typical cross-section  $IJKL$  is shown in the diagram.



- (i) Show, with working, that the cross-sectional area of the slice at  $h$  metres above the base, is given by

4

$$A = 8 - \frac{4h}{5} + \frac{h^2}{50}$$

- (ii) Hence by considering the typical slice  $IJKL$  of thickness  $\delta h$ , find the volume of the sculpture.

2

**End of paper**

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## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE :  $\ln x = \log_e x, \quad x > 0$



**Killara**  
HIGH SCHOOL

STUDENT NUMBER

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**Multiple Choice Answer Sheet**

Completely fill the response oval representing the most correct answer.

1.    A ○    B ○    C ○    D ○

2.    A ○    B ○    C ○    D ○

3.    A ○    B ○    C ○    D ○

4.    A ○    B ○    C ○    D ○

# Marking scheme

| MC        | 1. D    2. B    3. A    4. C                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                        |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q1<br>(i) | $y = \frac{\log_e x}{x}$ $y' = \frac{x \cdot \frac{1}{x} - \log_e x \cdot 1}{x^2}$ $= \frac{1 - \log_e x}{x^2}$                                                                                                                                                                                                                                                                                                                                                                   | 1 mark: correctly differentiates                                                                                                                                                                       |
| (ii)      | $\int_e^{e^2} \frac{1 - \log_e x}{x \log_e x} dx = \int_e^{e^2} \frac{1 - \log_e x}{x^2} \cdot \frac{x}{\log_e x} dx$ $= \int_e^{e^2} \frac{1 - \log_e x}{\frac{x^2}{\log_e x}} dx$ $= \left[ \log_e \left( \frac{\log_e x}{x} \right) \right]_e^{e^2}$ $= \log_e \left( \frac{\log_e e^2}{e^2} \right) - \log_e \left( \frac{\log_e e}{e} \right)$ $= \log_e \left( \frac{2 \log_e e}{e^2} \right) - \log_e \left( \frac{1}{e} \right)$ $= \log_e \left( \frac{2}{e} \right)$    | <p>1 mark: writes in <math>\frac{f'(x)}{f(x)}</math> form.</p> <p>1 mark: integrates correctly</p> <p>1 mark: correctly evaluates the integral.</p>                                                    |
| b)        | $\int_3^8 \frac{2x}{(x-1) + \sqrt{x+1}} dx$ <p>Let <math>u^2 = x+1</math><br/> <math>2u du = dx</math>      <math>x = 3, u = 2; \quad x = 8, u = 3</math> <b>1 mark</b></p> $\int_3^8 \frac{2x}{(x-1) + \sqrt{x+1}} dx = \int_2^3 \frac{2(u^2 - 1)2u du}{(u^2 - 2) + u}$ $= 4 \int_2^3 \frac{u^3 - u}{u^2 + u - 2} du \quad \mathbf{1 \text{ mark}}$ $= 4 \int_2^3 u - 1 + \frac{2u - 2}{u^2 + u - 2} du$ $= 4 \int_2^3 u - 1 + \frac{2}{(u+2)} du \quad \mathbf{1 \text{ mark}}$ | <p>1 mark: uses the correct substitution and converts the terminals.</p> <p>1 mark: correctly converts the integral in terms of <math>u</math>.</p> <p>simplifies the integrand into simpler terms</p> |

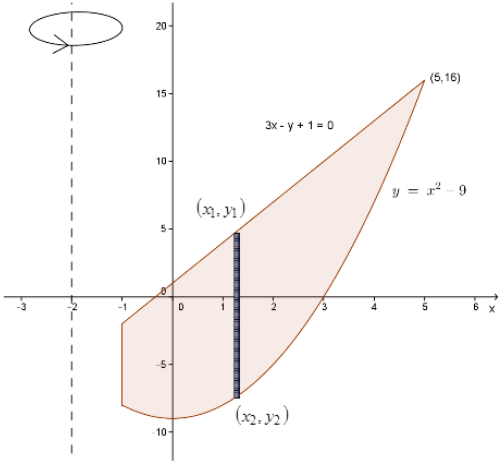
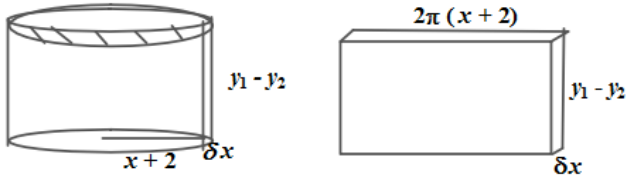
|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                             |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | $= 4 \left[ \frac{u^2}{2} - u + 2 \log_e(u+2) \right]_2^3$ $= 4 \left[ \left( \frac{9}{2} - 3 + 2 \log_e 5 \right) \right] - \left( \frac{4}{2} - 2 + 2 \log_e 4 \right)$ $= 6 + 8 \log_e \left( \frac{5}{4} \right) \quad \mathbf{1 \text{ mark}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1 mark: correct integration and evaluation.                                                                                                                                                                                                                 |
| c)(i) | $\frac{2x^2 + 7x - 1}{(x-2)(2x^2 + x + 1)} = \frac{A}{x-2} + \frac{Bx+C}{x^2 + x + 1}$ $A(x^2 + x + 1) + (Bx+C)(x-2) = 2x^2 + 7x - 1$ $x=2, \quad 7A=21 \quad \mathbf{1 \text{ mark}}$ $A=3$ <p>Coefficient of <math>x^2</math>, <math>2 = A + B</math></p> $B = -1$ <p>Constant, <math>-1 = A - 2C</math></p> $2C = 4 \quad \therefore C = 2 \quad \mathbf{1 \text{ mark}}$                                                                                                                                                                                                                                                                                                                                            | 1 mark: writes the equivalent polynomial to $2x^2 + 7x - 1$ in terms of A, B and C and attempts to evaluate A, B or C.<br><br>2 mark: Solves for the correct values of A, B and C.                                                                          |
| (ii)  | $\int \frac{2x^2 + 7x - 1}{(x-2)(2x^2 + x + 1)} dx = \int \frac{3}{x-2} + \frac{-x+2}{x^2 + x + 1} dx$ $= 3 \ln x-2  + \int \frac{-x+2}{x^2 + x + 1} dx \quad \mathbf{1 \text{ mark}}$ $= 3 \ln x-2  - \frac{1}{2} \int \frac{2x+1-5}{x^2 + x + 1} dx$ $=$ $3 \ln x-2  - \frac{1}{2} \int \frac{2x+1}{x^2 + x + 1} dx + \frac{5}{2} \int \frac{1}{\left(x + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} dx$ $\quad \mathbf{1 \text{ mark}}$ $= 3 \ln x-2  - \frac{1}{2} \ln x^2 + x + 1  + \frac{5}{2} \cdot \frac{2}{\sqrt{3}} \tan^{-1} \frac{2x+1}{\sqrt{3}} + C$ $= 3 \ln x-2  - \frac{1}{2} \ln x^2 + x + 1  + \frac{5}{\sqrt{3}} \tan^{-1} \frac{2x+1}{\sqrt{3}} + C \quad \mathbf{1 \text{ mark}}$ | 1 mark: expresses the integrand in $\frac{A}{x-2} + \frac{Bx+C}{x^2 + x + 1}$ form and integrates $\frac{A}{x-2}$ correctly.<br><br>1 mark: correctly separates $\int \frac{-x+2}{x^2 + x + 1} dx$ to integrable form.<br><br>1 mark: correctly integrates. |

## Question 2

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                              |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a)        | $t = \tan \frac{x}{2},$ <p>When <math>x = 0</math>, <math>t = 0</math>; <math>x = \frac{\pi}{2}</math>, <math>t = 1</math> <math>dx = \frac{2dt}{1+t^2}</math> <b>1 mark</b></p> $\int_0^{\frac{\pi}{2}} \frac{1}{2 \sin x + \cos x + 3} dx =$ $\int_0^1 \frac{1}{2 \cdot \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} + 3} \frac{2dt}{1+t^2}$ $= \int_0^1 \frac{\frac{2dt}{1+t^2}}{\frac{4t+1-t^2+3+3t^2}{1+t^2}}$ $= \int_0^1 \frac{2dt}{2t^2 + 4t + 4}$ $= \int_0^1 \frac{dt}{t^2 + 2t + 2} \quad \text{1 mark}$ $= \int_0^1 \frac{dt}{(t+1)^2 + 1}$ $= \left[ \tan^{-1}(t+1) \right]_0^1 = \tan^{-1}(2) - \tan^{-1}(1)$ $= \tan^{-1}(2) - \frac{\pi}{4} \text{ or } \tan^{-1}\left(\frac{1}{3}\right) \quad \text{1 mark}$ | <p>1 mark: converts the limits and <math>dx</math> in terms of <math>t</math></p> <p>1 mark: uses the correct substitutions for <math>\sin x</math> and <math>\cos x</math> and simplifies to give</p> $\int_0^1 \frac{dt}{t^2 + 2t + 2}$ <p>1 mark: correctly integrates and evaluates.</p> |
| b)<br>(i) | $I_n = \int_0^1 (1-x^2)^n dx$ $= \int_0^1 (1-x^2)^n \cdot 1 dx$ $= \left[ (1-x^2)^n \cdot x \right]_0^1 - \int_0^1 n(1-x^2)^{n-1} \cdot -2x \cdot x dx \quad \text{2 marks}$ $= 0 + 2n \int_0^1 x^2 (1-x^2)^{n-1} dx$ $= 2n J_{n-1} \quad \text{1 mark}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>1 mark (each part): applies integration by parts.</p> <p>1 mark: deduces the correct result.</p>                                                                                                                                                                                          |

|       |                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                             |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (ii)  | $I_n = 2n J_{n-1}$ $= 2n \int_0^1 x^2 (1-x^2)^{n-1} dx$ $= 2n \int_0^1 [1 - (1-x^2)] (1-x^2)^{n-1} dx \quad \mathbf{1 \text{ mark}}$ $= 2n \int_0^1 (1-x^2)^{n-1} dx - 2n \int_0^1 (1-x^2)^n dx$ $= 2n I_{n-1} - 2n I_n$ $(2n+1) I_n = 2n I_{n-1}$ $I_n = \frac{2n}{2n+1} I_{n-1} \quad \mathbf{1 \text{ mark}}$                                                                      | <p>1 mark: expresses <math>x^2</math> as <math>1 - (1-x^2)</math> and rewrites the integrand into difference of two terms.</p> <p>1 mark: uses the definition of <math>I_n</math> and <math>I_{n-1}</math> to give the required result.</p> |
| (iii) | $J_n = \int_0^1 x^2 (1-x^2)^n dx$ $\int_0^1 [1 - (1-x^2)] (1-x^2)^n dx$ $= \int_0^1 (1-x^2)^n dx - \int_0^1 (1-x^2)^{n+1} dx$ $\therefore J_n = I_n - I_{n+1} \quad \mathbf{1 \text{ mark}}$ $I_n - \frac{2(n+1)}{2(n+1)+1} I_n \text{ using the result from (ii)}$ $= I_n \left[ \frac{2n+3-2n-2}{2n+3} \right]$ $= I_n \left( \frac{1}{2n+3} \right) \quad \mathbf{1 \text{ mark}}$ | <p>1 mark: expresses <math>x^2</math> as <math>1 - (1-x^2)</math> to get the required result</p> <p>1 mark: uses the result from (ii) for <math>I_{n+1}</math> and simplifies to get the result</p>                                         |
| (iv)  | $J_n = \int_0^1 x^2 (1-x^2)^n dx$ $= \left( \frac{1}{2n+3} \right) I_n \text{ from (iii)}$ $= \left( \frac{1}{2n+3} \right) 2n J_{n-1} \text{ using (i)}$ $= \frac{2n}{2n+3} J_{n-1}.$                                                                                                                                                                                                | <p>1 mark: uses the results from (i) and (iii) to give the relationship.</p>                                                                                                                                                                |

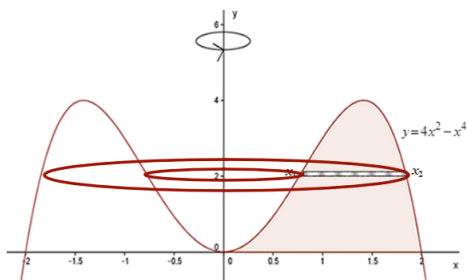
### Question 3

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>a)</p> <p>(i)</p> |  <p> <math>x^2 - 9 = 3x + 1</math><br/> <math>x^2 - 3x - 10 = 0</math><br/> <math>(x - 5)(x + 2) = 0</math><br/> <math>x = 5, -2</math> </p>  <p><b>1 mark</b></p> <p>Volume of the slice is <math>\delta V = 2\pi(x + 2)(y_1 - y_2)\delta x</math> <b>1 mark</b></p> $= 2\pi(x + 2)(3x + 1 - x^2 + 9)\delta x$ $= 2\pi(x + 2)(10 + 3x - x^2)\delta x$ <b>1 mark</b> | <p>1 mark: Draws the sketch of the cylindrical shell with correct dimensions</p> <p>1 mark: Develops the volume of the cylindrical shell <math>\delta V</math> by finding the volume of the rectangular prism ( diagram or explanation required)</p> <p>1 mark: converts <math>y_1 - y_2</math> in terms of <math>x</math> and proves the required expression for <math>\delta V</math></p> <p>(ii)</p> <p>1 mark: writes the expression for <math>V</math> as the sum of <math>\delta V</math> s and let <math>\delta x \rightarrow 0</math> And gives the form</p> $2\pi \int_{-1}^5 (x + 2)(10 + 3x - x^2) dx$ <p>1 mark: correctly integrates to give the correct answer.</p> |
|                      | <p>Summing up the volumes of all such typical shells between <math>x = -1</math> to <math>x = 5</math> and then let <math>\delta x \rightarrow 0</math> gives the volume of the solid.</p> <p>Thus,</p> $V = 2\pi \int_{-1}^5 (x + 2)(10 + 3x - x^2) dx$ <b>1 mark</b> $= 2\pi \int_{-1}^5 (10x + 3x^2 - x^3 + 20 + 6x - 2x^2) dx$ $= 2\pi \int_{-1}^5 (20 + 16x + x^2 - x^3) dx$ $= 2\pi \left[ 20x + 8x^2 + \frac{x^3}{3} - \frac{x^4}{4} \right]_{-1}^5$                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

$$= 2\pi \left[ 100 + 200 + \frac{125}{3} - \frac{625}{4} \right] - \left[ 20 + 8 + \frac{1}{3} - \frac{1}{4} \right]$$

$$= 396\pi \text{ unit}^3. \quad \mathbf{1 \text{ mark}}$$

b)



Area of the Annulus  $\delta A = \pi(x_2^2 - x_1^2)$

Volume of the slice  $\delta V = \pi(x_2^2 - x_1^2)\delta y$  **1 mark**

$y = 4x^2 - x^4$

ie.  $x^4 - 4x^2 + y = 0$

$$x^2 = \frac{4 \pm \sqrt{16 - 4y}}{2} = 2 \pm \sqrt{4 - y}$$

$$x_1^2 = 2 - \sqrt{4 - y} \quad x_2^2 = 2 + \sqrt{4 - y}$$

$$\therefore x_2^2 - x_1^2 = 2\sqrt{4 - y}$$

$$\therefore \delta V = \pi(2\sqrt{4 - y})\delta y \quad \mathbf{1 \text{ mark}}$$

Volume of the solid =  $V = 2\pi \int_0^4 \sqrt{4 - y} \delta y$

$$= -2\pi \cdot \frac{2}{3} \left[ (4 - y)^{\frac{3}{2}} \right]_0^4$$

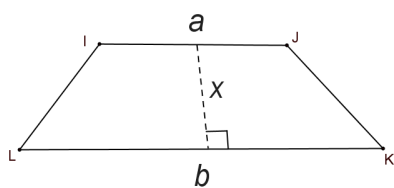
$$= \frac{-4\pi}{3} (0 - 8) = \frac{32\pi}{3} \text{ unit}^3 \quad \mathbf{1 \text{ mark}}$$

1 mark: gives the cross-section area of the perpendicular slice.

1 mark: gives the volume of the slice in terms of  $y$ .

1 mark: calculates the volume

c)  
(i)



When  $h = 0$ ,  $a = 0$   
 $h = 10$ ,  $a = 2$

4 marks: Expresses  $a$ ,  $b$  and  $x$  in terms of  $h$  and calculates the area of the cross-section  $IJKL$ .

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                           |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | $a = mh + p$ $h = 0, \quad a = 0$ $a = mh$ $= \frac{2-0}{10-0}h \quad \therefore a = \frac{h}{5}$ <p>Similarly,</p> <p>When <math>h = 0, \quad b = 4</math><br/> <math>h = 10, \quad b = 0</math></p> $\therefore b = \frac{0-4}{10-0}h + 4 \quad b = \frac{-2}{5}h + 4$ <p>Also,<br/> When <math>h = 0, \quad x = 4</math><br/> <math>h = 10, \quad x = 2</math></p> $\therefore x = \frac{2-4}{10-0}h + 4 \quad x = \frac{-1}{5}h + 4$ <p>Area of <math>IJKL = \frac{x}{2}(a+b)</math></p> $= \frac{1}{2} \left( \frac{-1}{5}h + 4 \right) \left( \frac{h}{5} + \frac{-2}{5}h + 4 \right)$ $= \frac{h^2}{50} - \frac{2h}{5} - \frac{2h}{5} + 8$ $= \frac{h^2}{50} - \frac{4h}{5} + 8$ | <p>Subtract 1 mark for errors in each of the calculations.</p> <p>Award 1 mark: If an attempt to find the area of cross-section with reasonable mathematical working.</p> |
| (ii) | <p>Volume of the cross-section <math>\delta V = Ah</math></p> $= \left( \frac{h^2}{50} - \frac{4h}{5} + 8 \right) \delta h$ <p>Summing up the volumes of all such typical slices between <math>h = 0</math> to <math>h = 10</math> and then let <math>\delta h \rightarrow 0</math> gives the volume of the solid.</p> <p>Thus volume of the solid</p> $V = \int_0^{10} \left( \frac{h^2}{50} - \frac{4h}{5} + 8 \right) dh \quad \text{1 mark}$ $\left[ \frac{h^3}{150} - \frac{4h^2}{10} + 8h \right]_0^{10}$ $= \frac{140}{3} - 0 = \frac{140}{3} \text{ unit}^3 \quad \text{1 mark}$                                                                                                | <p>1 mark: develops the correct expression for <math>V</math>.</p> <p>1 mark: Calculates the correct volume.</p>                                                          |